Prime Services
Capital Solutions
Hedge Fund Pulse

BARCLAYS

The Curious Case of Converts

Recent Developments in the Convertible Hedge Fund Landscape

November 2013

For institutional and professional investors only. Not for retail investors and not for redistribution. This document is not research. This document is produced by Barclays Capital Solutions Group, not Barclays Research Department. This document does not constitute legal, business, investment, accounting or tax advice. See additional disclaimer on back cover.

Contents

I.	Study overview	1
II.	Executive summary	1
III.	Current landscape	2
IV.	Changes in the convert marketplace	5
V.	Investment considerations	11
VI.	Conclusions	14
VII.	Acknowledgments	15
VIII.	Capital Solutions	15

I. Study overview

For this edition of the HF Pulse, the Strategic Consulting team decided to explore the convertible (convert) Hedge Fund (HF) landscape. Despite the relatively small AUM base of HF strategies focussed on converts, the team found the subject matter intriguing for two main reasons: One, while a number of HF strategies performed poorly in 2008 and suffered heavy redemptions, most appear to have recovered and are back in favour with investors (e.g., mortgages, structured credit) but this appears not to be the case for convert HFs. Despite the fact that, on average, convert HFs have produced some of the most compelling returns among key HF strategies since 2008, they have fared the worst in terms of net flows into the strategy[1]. Since flows generally track performance over time, this conspicuous inconsistency is surprising, especially in the current return-seeking environment. Two, the convert market itself has undergone significant changes since 2008, both on the supply side (e.g., market size, ownership, leverage, liquidity) and the demand side (e.g., convert HF AUM, number of players, strategy). In a number of ways, convert HFs today are very different from what they were pre-crisis. In this study, we take a closer look at these changes and their implications with an eye toward helping investors better assess whether the strategy should play a role in their HF portfolios.

Outlined below are the main questions we have addressed across three key topics in this study:

I. Current landscape for convert HFs

- What happened to convert HFs in 2008?
- What has been the impact on the strategy's AUM and number of funds?
- How has the strategy performed since 2008 and what has been the related investor sentiment / reaction?

II. Changes in convert market

- How has the convert landscape changed since 2008? In particular, what have been the:
  1. Supply side changes – e.g., market size, level of new issuance, ownership structure, liquidity, and leverage
  2. Demand side changes – e.g., changes in the convert HF strategy and talent base

III. Investment considerations

- How correlated is convert valuation with net returns of convert HFs?
- How do the risk / return profiles of long-only and convert HF products differ?
- What are the pros and cons of allocating to convert HFs through pure-play convert HFs versus multi-strategy (MS) HFs that also invest in converts?
- In general, what are the key attributes investors embrace in a convert HF? What do they avoid?

Methodology

To address the above questions, we tapped three sources to gather required information for analysis:

1. In-depth interviews with 15 convert HF managers / investors
   - Ten convert HF managers (including five dedicated convert HFs and five multi-strategy HFs with allocations / funds dedicated to converts)
   - Five HF allocators / consultants
2. Multiple interviews / discussions with Barclays internal convert specialists
   - Convert research
   - Converts sales and trading desks
3. External research and analysis
   - Commercial HF databases such as HFR and BarclayHedge
   - Industry reports and press articles

II. Executive summary[2]

The following are the high level takeaways from the study.

I. Current convert landscape

- **AUM / flows / performance** – A confluence of market, technical, and sector-specific factors led to a perfect storm in the convert market in 2008. Since then, convert HF AUM growth has slowed due to heavy redemptions despite strong performance.
  - Generally speaking, investor flows tend to track fund returns over time, but this has not been the case for convert HFs. Since 2008, compared to key HF strategies, the convert HF strategy has produced an annualised return of 5% (one of the best), and yet has experienced net outflows of ~25% (one of the worst). Additionally, the number of convert funds has fallen by ~45% since the 2007 peak (from 152 to 84).
- **Investor sentiment** – According to our survey participants, two contributing factors to this performance versus flow inconsistency include:
  - Investors that were 'traumatised' by 2008 events and their impact on convert HFs generally do not regret missing out on recent strong performance.
  - Investors' misconceptions about the strategy, and / or a lack of awareness about recent changes in the space.

II. Changes in the convert market

The convert market has undergone significant changes since 2008:

- **Market size / new issuance** – Total market size and new issuance have steadily declined post-crisis.
  - Global convert market has declined from $465bn (face value) in size at the end of 2008 to $315bn YTD 2013, a reduction of 30%+.

[1] Strategic Consulting analysis of HFR Convertible Arbitrage Index [2] All figures cited in executive summary are sourced in main document as they are discussed in detail

- Issuance hit lows of $48bn in 2011 and $55bn in 2012, though annualised YTD 2013 issuance is much stronger (~$70bn annualised).
- Interestingly, convert HF returns have held up during this time. Our analysis seems to suggest that market size / new issuance is only one of many factors impacting returns (correlation between market size and convert HF returns, as well as between issuance and returns has been relatively weak).

- **Ownership** – A prominent change since 2008 in the convert space has been the ascendency of the long-only convert buyers (outrights) as a dominant category of player.
  - The ownership split between HFs versus outrights in the US converts market was ~75% / 25% pre-2008. Today the split is roughly 50% / 50%. The shift in the non-US convert market is likely to have been even more pronounced in favour of outrights to ~40% / 60%.
  - The shift in the buyer base has prompted structural changes to the convert market in terms of

    i. Liquidity – Fewer HFs and the elimination of proprietary trading teams at banks mean less liquidity although, relative to its market size, liquidity has held up better than the high yield (HY) market.

    ii. Leverage – We estimate that the leverage in the convert market today is roughly one-third of the pre-2008 level.

    iii. Opportunity for return generation – More diversified / balanced ownership of converts is likely to create a broader opportunity set and less concentrated trades.

- **Strategy** – There are three main sources of returns for convert HFs: relative value, arbitrage and opportunistic trading. A majority of convert HFs in the present environment focus more on relative value (RV) opportunities. Convert HFs today have become more broad-based, fundamentals-driven, and catalyst-driven.

- **Talent base** – In part due to massive consolidation in the convert HF space, surviving convert HFs today tend to comprise a more experienced group of managers, on average, compared to other HF strategies evidenced by:
  - An average age of 7.4 years for existing convert HFs versus 6.3 years for all other HF strategies.
  - Convert HFs tending to possess a broader skill set than they did historically, according to our surveyed participants.

III. Investment considerations

- **Entry point** – A common concern for investors considering allocations to convert HFs in the current environment is that the asset class has been 'fully priced', i.e., converts are trading in line with or 'richer' than valuations. Our analysis, however, suggests that in 'normal' markets, convert HF returns are not driven primarily by the theoretical 'cheapness' of the market. That is just one of many factors that impact returns.

- **Long-only versus HFs** – Convert HFs have historically generated a 'smoother' return profile and a higher return on a risk-adjusted basis than long-only managers.

- **Multi-strategy versus pure-play convert HFs** – Pure-play and multi-strategy HFs approach the convert opportunity differently in terms of (1) strategy focus – MS focusses mostly on opportunistic plays (or convert beta) versus pure-play firms that also strive to capture relative value and arbitrage opportunities, and (2) allocation levels – MS allocations to converts fluctuate substantially depending on market environment (e.g., ~$70bn in 2009 versus ~$20bn today).
  - Key considerations for investors include their (1) desired strategy exposure, (2) risk tolerance, and (3) number of managers in their HF portfolio.
  - Since 2008, 'pure-play' HFs have demonstrated a performance edge versus MS (30% cumulative return for convert HFs versus 18% for MS over the 2008 to 2012 period). Most of the outperformance was generated in 2009, where pure-play convert HFs were able to move down the risk curve faster and more aggressively.

- **Manager selection** – Today, investors appear to prefer convert HFs with a broad-based skill set, fundamentals-driven approach, and an active trading style.

III. Current landscape

Recap of 2008 events

We would be remiss if we started discussing where convert HFs are today without mentioning the 2008 financial crisis, the watershed event from which many convert HFs are still recovering. While most asset classes struggled during that year amid a developing credit crisis, converts in particular experienced unprecedented deterioration in valuations. Due to this and other factors, the strategy returned -33%, versus -18% for the average HF and -38% for the S&P 500 index. Figure 1 summarises the confluence of market, technical, and convert-specific factors that led to this perfect storm, including:

- **Market:** Sharp decline in equity markets, dramatic widening in credit spreads, and a collapse in the valuation of convert preferreds.
- **Technical:** Ban on short selling, redemption-fuelled selling pressure, significant investor and broker dealer deleveraging, and a dramatic increase in risk aversion.
- **Convert-specific:** Crowded market dominated by HFs and proprietary desks, excessive leverage in system, and dramatically higher funding costs.

AUM and number of funds

All of the above led to heavy redemptions and massive consolidation in the convert HF space. To provide some historical context, we depict in Figure 2 the evolution of convert HF AUM since 1990. Before 2000, convert HF AUM grew at a rate of ~40%+ CAGR (though starting from a small base). This was the 'golden era' of convert HFs when only a select few played in the field. Average annualised returns were 10%+ during this time (not shown); net flows accounted for ~70% of total AUM growth of convert HFs. Between 2001 and 2007, convert HF AUM grew at a slower, but still healthy, CAGR of ~12% (despite a sell-off in 2005), and the growth during this time was driven equally by flows and performance. In this time period, the convert HF space became more crowded and the use of leverage rose. Since 2008, the chart shows a notable slowdown in convert HF AUM growth, (~1% CAGR) – the net effect of significant outflows (-$10bn) almost offsetting strong performance gains ($13bn).

Similarly, Figure 3 compares the changes in the number of funds in the overall HF industry[3] (top chart) versus the number of convert HFs[4] (bottom chart). Since 2007, the number of convert HFs has fallen from 152 to 84 (-45%) while the overall HFs in the industry have increased from 7,634 to 8,167 (+7%). It is interesting to note that between 2007 and 2008, the number of convert HFs actually decreased proportionally with the total number of HFs in the industry (both declined by ~10 – 15%); however, the overall industry has rebounded in subsequent years (+19% from 2008 to 1H 2013) while convert HFs have continued to decline in number (by an additional -36%).

[3] HFR, 2007 to 2013 1H [4] Figures represent the total number of funds that list "convertible arbitrage" as the primary strategy in HFR database from 2007 to 2013 1H. Please note that the figures represent number of funds, not firms. By our estimate, there are ~50 convertible firms in 2013 1H

FIGURE 1: Recap of 2008 Events and Impact on Convert HFs

2008 Performance	
Converts	
Long-Only[1]	-36%
Convertible HFs[2]	**-33%**
Key Indices	
S&P 500	**-38%**
10-Year Treasury	20%
Barclays US Credit	-4%
Barclays US HY	-32%
HF Strategies[3]	
Total HFs	**-18%**
Equity Hedge	-27%
Global Macro	5%
Event Driven	-22%
Relative Value	-18%

Key Reasons for Convert HF Underperformance

Market
- **Sharp decline in equity markets** – S&P 500 down 38%
- **Dramatic widening in credit spreads** – HY index OAS increased to ~2,000 bps in Oct 2008 from ~400 bps one year before[4]
- **Collapse in the valuation of convertible preferreds** – significant value destroyed in Financials (sector with nearly two-thirds of total new issuance in 2008)[4]

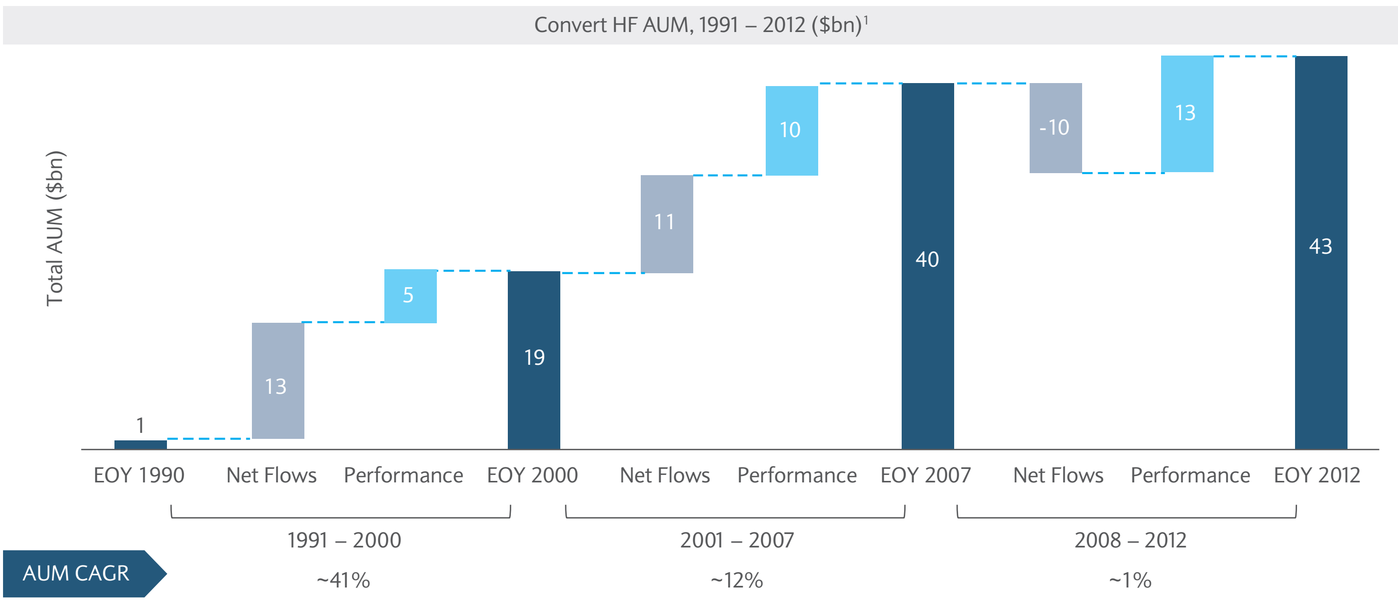
Technical
- **Ban on short selling** – severely impaired Convert HFs' ability to hedge
- **Redemption-fuelled selling pressure** – net outflow of ~36% during 2008 / 2009 from Convertible HFs[2]
- **Significant investor and broker dealer deleveraging** – dramatic increase in risk aversion; flight to quality[4]
- **Scarce liquidity** – trading volume declined to ~$30bn in 11/08 from ~$60bn in 01/08, driven by investor reduction, deleveraging, broker balance sheet (b/s) contraction[4]

Convert Specific
- **Crowded market dominated by HFs and proprietary desks** – 75% / 25% split of convert ownership between HFs and outrights[5]
- **Excessive leverage in system** – ~4 – 5x versus 1 – 2x today[6]
- **Higher funding cost** – many broker dealers stopped accepting convertibles as collaterals

Gating / redemption suspension | Mass exodus from Convert HFs | Consolidation among Convert HFs

1. Barclays Convertibles Research estimates 2. HFRI Convertible Arbitrage Index 3. HFRI Indices 4. Barclays Convertibles Research, "Rising from the Ashes", December 2008
5. Barclays Convertibles Research, "Ready to Bat", December 2012 6. Strategic Consulting estimates

FIGURE 2: Convert HF AUM Growth



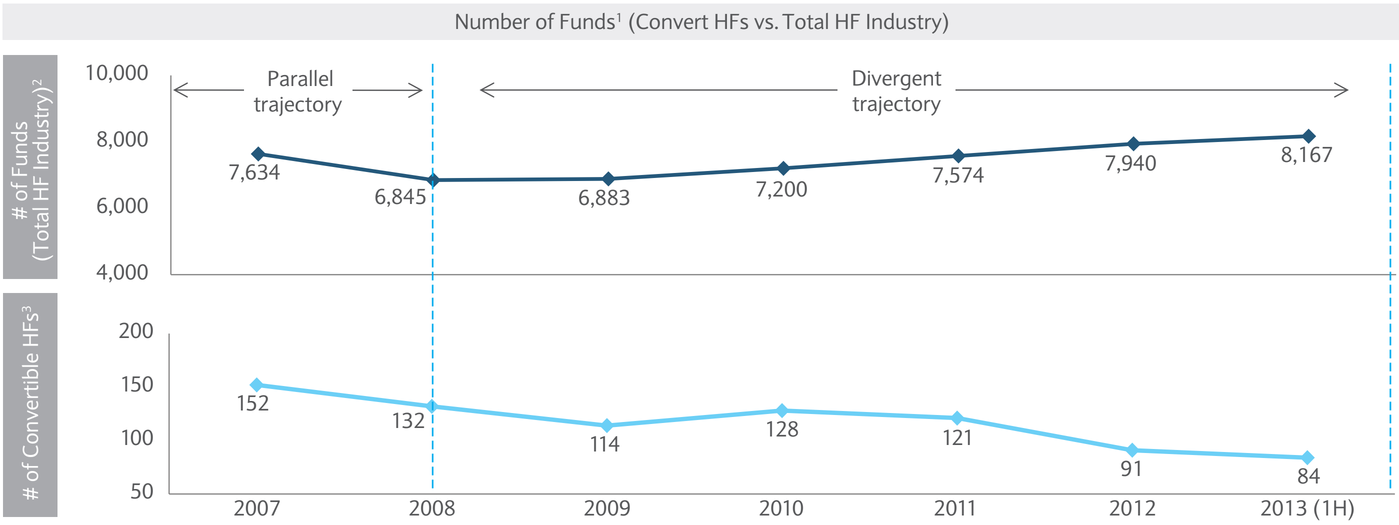
1. HFRI Convertible Arbitrage Index

Performance and flows

So how has the convert HF strategy performed amid all the negative sentiment and challenges in the past five years? Interestingly, the strategy has produced some of the most compelling returns since the 2008 events. Figure 4 shows five-year annualised returns for key HF strategies (x-axis) against the related five-year cumulative net flows as a percent of their respective AUM (y-axis). Generally speaking, investor flows have tended to track fund returns, i.e., strategies that have performed better have also attracted more flows over time (as evidenced by the fact that most data points fall relatively close to the dotted line). However, convert HFs appear to be a strong outlier in this respect. The strategy has produced an annualised return of ~5% since 2008 (one of the best) yet has experienced net outflows of 25% (one of the worst). Macro strategies, on the other hand, have seen more flows than would be predicted by their performance. We are a bit surprised to find such conspicuous inconsistency, especially in today's return-seeking environment. The right side of Figure 4 displays select manager / investor quotes that are illustrative of their thinking about this issue. In summary, we find two key contributors to this performance / flow discrepancy:

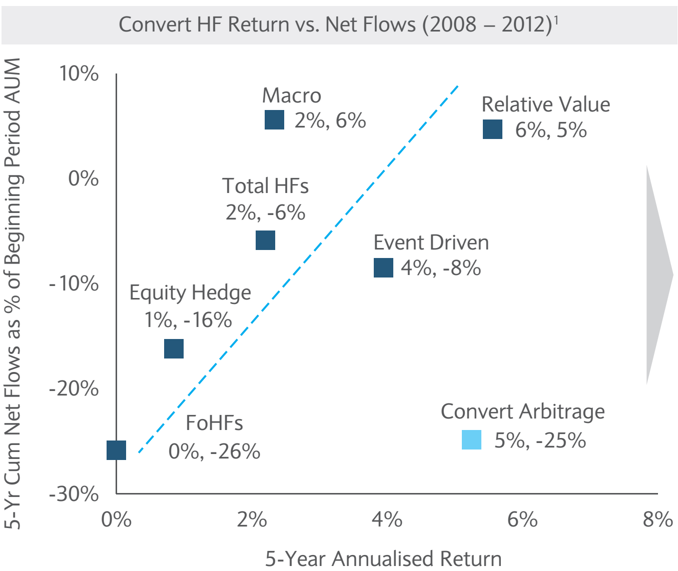
- Some investors are still 'traumatised' by 2008 events and, as a result, have no regrets about missing out on recent strong performance.

FIGURE 3: Number of Convert Funds



1. Please note, the figures shown in this chart represent number of funds, not firms 2. HFR 3. HFR Figures represent the total number of funds that list "convertible arbitrage" as the primary strategy in HFR database from 2007 to 2013 1H. Please note that the figures represent number of funds, not firms. By our estimate, there are ~50 convert firms in 2013 1H.

FIGURE 4: Performance vs. Net Flows since 2008



Manager / Investor Feedback

"The negative bias seems to be an emotional reaction and, in our view, often unsubstantiated." – Convert HF

"We had a painful experience in '08. I'd rather miss a couple years of good performance than repeat that experience." – HF Investor

"There are a number of misconceptions about the space on multiple levels." – Convert HF

"Many investors are not aware of the many structural changes that have taken place since 2008." – Convert HF

"I wish I was more informed. We have not had dedicated coverage of the space since 2008." – HF Investor

1. HFRI Convertible Arbitrage Index

- Investors' misconceptions about the strategy, and / or lack of awareness about recent changes in the space.

We examine some of the changes in the convert marketplace in the next section.

IV. Changes in the convert marketplace

Changes in market size and issuance

As of the end of September 2013, there were ~$315bn of converts outstanding globally[5] across ~800 securities / issuers.[6] The size of the global convert market, as depicted in the left chart of Figure 5, has declined post-crisis at a CAGR of -7.6%. On a relative basis, EMEA and APAC ex-Japan markets have held up better than US and Japan. In 2008, the geographic distribution of converts was roughly 62% US, ~20% EMEA, 10% APAC ex-Japan and 8% Japan, versus 55% US, ~30% EMEA, ~10% APAC ex-Japan and 5% Japan today. This trend toward increased geographical diversity may provide incremental opportunities for those convert HFs that have a global mandate.

In line with market size, the level of convert issuance has steadily decreased, reaching a low of $48bn in 2011 (followed by $55bn in 2012), as shown on the right side of Figure 5. Key factors that have contributed to declining market size / issuance may include (1) declining treasury rates and all-in yields for HY and Investment Grade (IG) bonds have made tapping these markets compellingly more attractive than converts, (2) low stock prices and cheap equity valuation have made issuers hesitant to issue converts, and (3) a lackluster economic environment has led issuers to be more defensive and less expansionary with their plans[7].

As shown in Figure 5, in 2013, however, issuance has seen a notable pickup (~$70bn annualised) from prior year lows, which could herald a turnaround (although net issuance is still projected to be negative). Issuance, like any other form of capital raising, is cyclical and can be influenced by a large set of macro variables. Listed below are the major trends / factors that we believe support an increase in convert issuance in the current environment[8]:

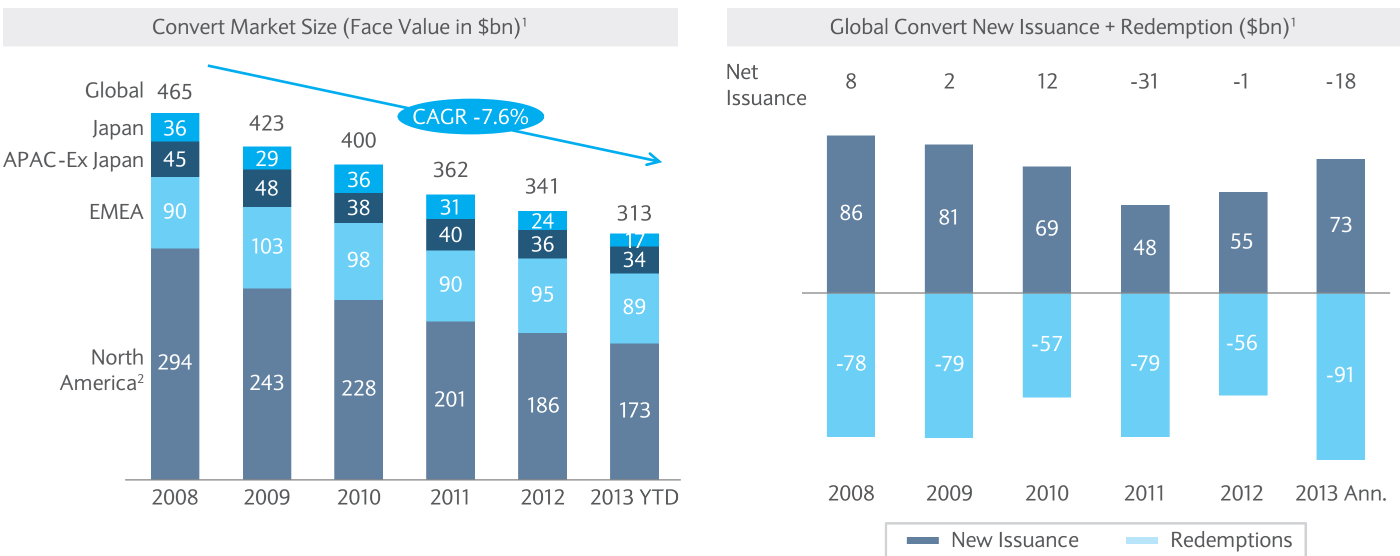
- Increasing rates and credit yields.
- Rising equity market and strengthening equity valuations should make issuers more willing to issue converts.
- If equity volatility picks up, it would result in potentially better terms for issuers.
- Need for growth / expansionary capital is likely to grow.

Nonetheless, the falling market size and issuance levels in recent years have been a concern to some investors. Many believe that the convert space is an issuance driven market, and declining issuance portends a dwindling opportunity set and, ultimately, poor returns. What is surprising, however, is that, convert HF performance has held up well during this period, which leads us to think that convert HF returns may be less dependent on market size / issuance than some investors believe. As shown in the chart on the left in Figure 6, the six-year correlations between convert market size / issuance and convert performance (both long-only and HF) have been relatively weak, i.e., between -0.3 and 0.3.

This is not to say that market size / issuance levels are not important. We do believe that new issuance represents a key source of incremental opportunity for convert HFs. Additionally, studies have shown that new issues, on average, are launched at a 2 – 3% discount to theoretical fair value, and

[5] At face value [6] Barclays Convertibles Research estimates. Figures do not include (1) convert issues with less than $50mn face value in US market and with less than $100mn in non-US market, (2) defaulted securities, (3) strictly private placements and (4) Canadian and Latin American converts [7,8] "As Good as It Gets: An Opportune Time for Convert Issuance", Barclays Convertibles Research, June 2013

FIGURE 5: Market Size, Issuance and Redemptions



1. Barclays Convertibles Research estimates. Figures do not include (1) convert issues less than $50mn for US and less than $100mn for non-US, (2) defaulted securities, (3) strictly private placements and (4) Canadian and Latin American converts 2. Predominantly US; Does not include Canadian converts

about two-thirds of this 'cheapness' is quickly captured by investors[9]. What our analysis does suggest, however, is that market size / issuance has not driven convert HF performance because it appears to be only one of many factors at play. One such factor, for example, is the concentration of players. As illustrated in the chart on the right in Figure 6, while the size of global convert market has shrunk since 2008 (x-axis), it has fallen in proportion to the decrease in the number of convert HFs (y-axis), therefore the 'opportunity pie per convert HF' has theoretically remained the same.

Changes in ownership

One of the most significant structural changes in the convert space post-crisis has been the ascendency of the long-only convert buyers (outrights) as a dominant category of player. Before 2008, the ownership split between HFs versus outrights in the US convert markets was ~75% versus 25%, respectively, as illustrated in Figure 7. Today, the split is ~50% / 50%[10]. The shift has been even more pronounced for the non-US convert market. According to the Barclays Convertibles Research team, the mix in these markets is tilted in favour of outrights today

[9] Barclays Convertibles Research estimates [10] Barclays Convertibles Research, estimates based on data reported on firm 13F filings. Sample size represents ~70% of outstanding US convert securities

FIGURE 6: Correlation between Returns and Market Size / Issuance

Correlations between Returns and Market Size / Issuance[1]

Period		Long-Only Returns	HF Returns
2006 – 2012	Market Size	-0.3	-0.2
	New Issuance	-0.2	-0.2
2006 – 2012 Excluding '08 and '09	Market Size	-0.1	0.2
	New Issuance	-0.1	0.1

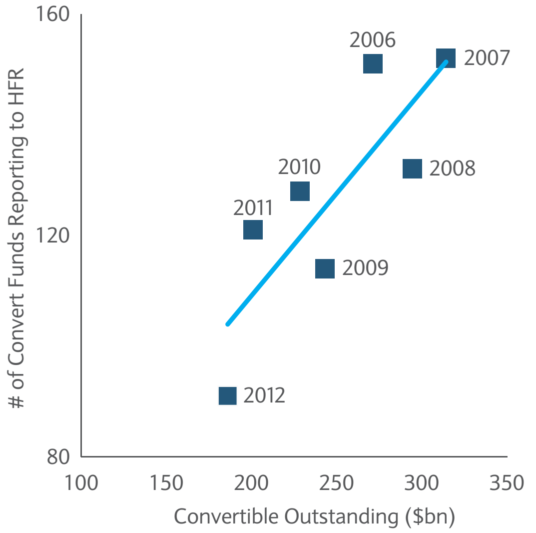
Manager Feedback

"It's a common misconception that issuance drives how well the market does."

"The actual return generated from the first day of the issuance is a very small part of the return."

"Investors may be overly concerned with issuance. It is one of the many factors affecting the market."

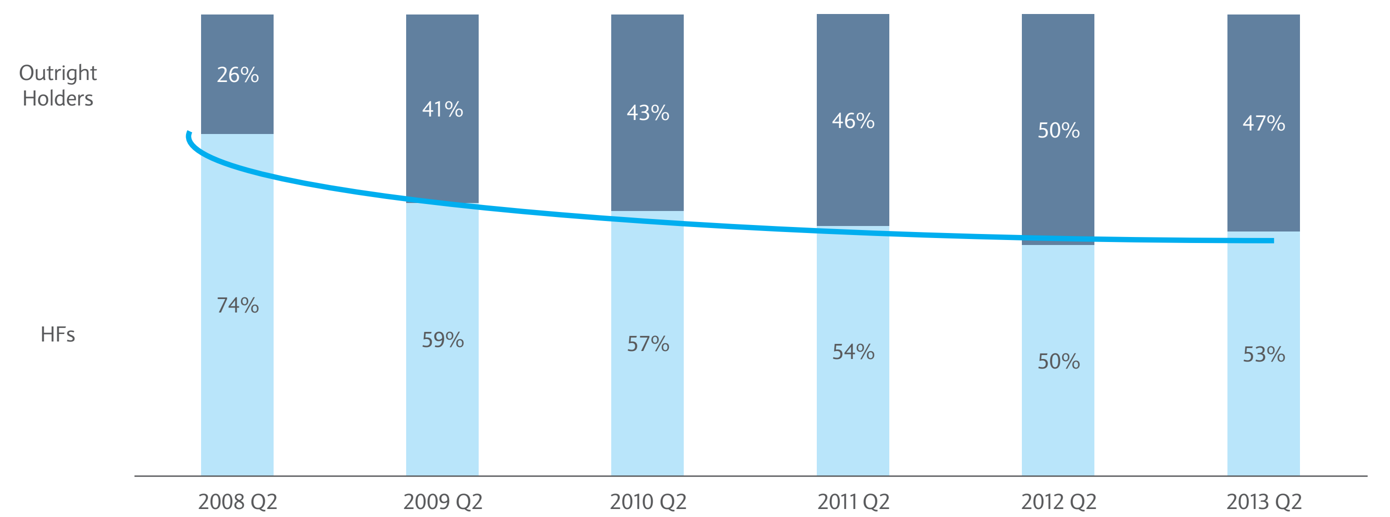
Convert Market Size Relative to # of Convert HFs



1. Strategic Consulting analysis. Market size, issuance, and long only return data used in analysis are US data based on Barclays Convertibles Research estimates. HF return data is the average annual return of HFR Convertible Arbitrage Index and BarclayHedge Convertible Arbitrage Index

FIGURE 7: Changes in Ownership

US Convert Ownership (2008 – 2013)[1]



1. Barclays Convertibles Research estimates, based on data reported on firm 13F filings. Sample size represents ~70% of outstanding securities

(i.e., ~55% – 65% outright). On a global basis, we estimate convert ownership to be roughly 40% HFs and 60% outrights today (not shown).

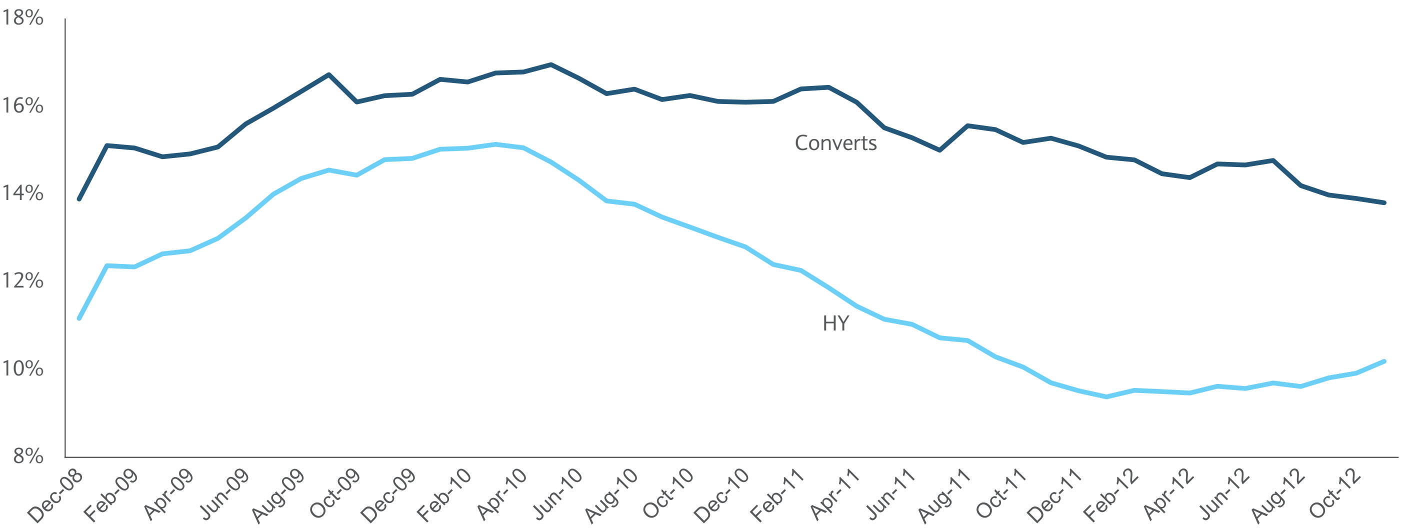
This shift in buyer base has prompted structural changes to the convert market in terms of its liquidity, leverage, and opportunities for return generation:

- Liquidity: Lower HF participation and elimination of prop teams mean less liquidity in the convert market as HFs and bank prop teams tend to trade more frequently than outrights.
  – Average monthly TRACE volumes have fallen from ~$32bn in 2009 to $20bn in 2012[11].
  – Additionally, factors like increased regulatory uncertainty, more modest leverage deployed, tepid new issuance (small deal sizes) and a shrinking convert market has put further pressure on secondary market liquidity[12].
  – However, relative to the size of market, liquidity in converts still appears relatively healthy. Figure 8 shows that TRACE volumes in the convert market, measured as a percentage of outstanding face value, have consistently been notably better than High Yield (HY).

[11] Bloomberg [12] Barclays Convertibles Research, "Ready to Bat", December 2012

FIGURE 8: Changes in Liquidity

Average Monthly[1] TRACE Volumes as a % of Face Outstanding for HY and Converts[2]



1. Average of last twelve months 2. Barclays Convertibles Research, "Ready to Bat – US Convertibles Outlook 2013", December 2012

FIGURE 9: Changes in Leverage

Typical Long Leverage Held by Convert HFs[1]

Estimated Leverage in Convert Marketplace (Including Outrights)[2]

1. Estimates based on Strategic Consulting analysis, Barclays Convertibles Research estimates, Strategic Consulting manager interview feedback 2. Derived from estimated long leverage held by convert HFs (chart on left) and convert ownership split by HFs and outright holders (Figure 7)

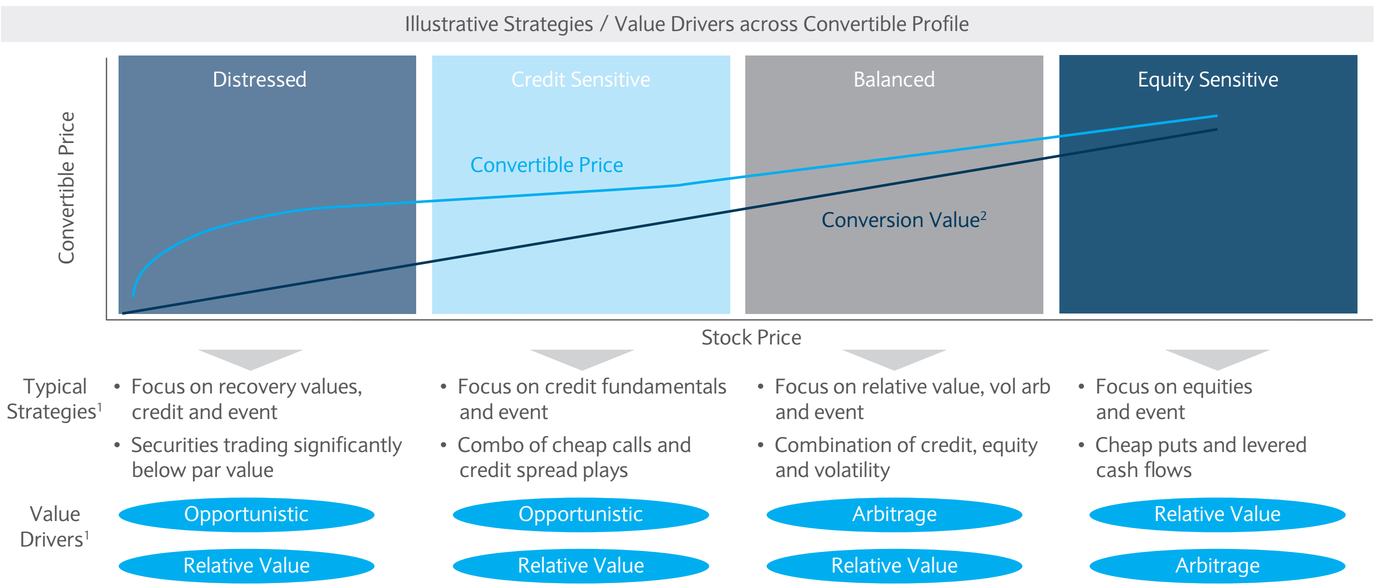
- Leverage: Fewer convert HFs results in reduced overall leverage in the convert market, which in turn reduces systemic risk in the space.
  - Based on our analysis and interviews with external and internal participants, we estimate that the long leverage level of convert HFs has fallen by ~60% from 2008 levels (from ~6x to ~2.5x), as shown in the chart on the left in Figure 9. It appears that the drop in leverage is the result of three key factors:
    - Selection bias – Convert HFs that resorted to excessively high leverage are likely to have closed after 2008.
    - Change in availability of financing – In general, it has become more challenging for convert HFs to obtain the same level of leverage post-2008.
    - Shift in approach / strategy – By and large, the convert strategy has become more directional and fundamentally driven, which has reduced the need for reliance on high leverage for returns.
  - Leverage in the system has fallen even more aggressively, largely due to the decline of HF participants versus outright buyers (who use no leverage), shown in the chart on the right in Figure 9. We estimate that leverage in the convert market today is roughly a third of the level in 2008.
- Opportunity for return generation: Because outrights and HFs have different investment objectives, a more diversified / balanced ownership base creates a broader opportunity set and less concentrated positions.
  - Outrights usually take a buy-and-hold approach and have a medium- to long-term investment horizon (six months to two years). They tend to be more yield-focussed and generally take a fundamental view with an equity bias. HFs, on the other hand, typically trade more actively and have a short to medium investment horizon (days to months). They tend to take a relative value approach using multiple investment strategies such as volatility arbitrage, credit, and cash flow / carry trades.
  - Furthermore, according to the managers we interviewed, arbitrage opportunities tend to persist longer today than they did historically as a result of lower HF participation in the market. The flip side is that it also takes longer for mispricings to converge to their true value, as valuation has become more vulnerable to technical factors.
  - Finally, a few convert managers told us that they feel more comfortable taking bigger short positions now than they did in the past because they recognise there are more stable longs (outrights) in the market than a few years ago.

Changes in strategy

In this section, we explore three key aspects of convert HF strategies: (1) Sources of opportunity, (2) Common sub-strategies / trades, and (3) Evolution of convert HF strategies over time.

Before delving into these topics, we thought it may be helpful to provide an overview of convert bond structure and valuation. A convert bond, in its simplest form, is a corporate bond with an embedded call option that allows the owner to convert the bond into a predetermined number of shares of the issuing company (i.e., conversion ratio). After a convert bond is issued, its price fluctuates depending on the relationship between the stock price and the conversion price. As illustrated in Figure 10, the relationship (or convert profile) can be described as 'equity-sensitive', 'balanced', 'credit-sensitive', or 'distressed'.

FIGURE 10: **Changes in Convert HF Strategy (I / III)**



1. Strategic Consulting manager interview feedback 2. Conversion value is the product of conversion ratio and stock price

The convert profile is considered:

- 'Equity-sensitive' when the stock price exceeds the conversion price (i.e., the embedded option is in-the-money). As the stock rises, the price of the convert becomes more sensitive to changes in the stock price and less so to changes in interest rates.
- 'Balanced' when the stock price is close to the conversion price (i.e., the embedded option is at-the-money). At this stage, the convert value is affected by both interest rates and the stock price.
- 'Credit-sensitive' when the stock price is less than the conversion price (i.e., the embedded option is out-of-the-money). At this stage, the convert value is influenced more by factors affecting its bond value, i.e., interest rates and credit factors, than the stock price. As the stock price declines, the convert price will fall but at a slower rate.
- 'Distressed' when the stock price declines and approaches zero (i.e., the embedded option is deep-out-of-the-money). At this stage, the company's credit is very distressed and the convert declines to its liquidation value.

1. Sources of opportunity for convert HFs

So how do convert HF managers typically generate returns? According to our interview participants, there are three main sources of opportunity convert HFs tap:

1. Relative Value

- Idiosyncratic opportunities including (1) relative mis-pricing of securities within the same company, (2) relative mis-pricing of securities with similar characteristics across companies, and (3) event driven trades that capitalise on the relationship between companies involved in a corporate event.
- These opportunities exist through all phases of the convert security life cycle.
- They typically require deep fundamental work on underlying firms.

2. Arbitrage

- Classic 'textbook' convert arbitrage strategy where the bond is trading at-the-money.
- Exploits cheap embedded volatility through active management of equity delta hedge.
- Tends to be more 'model' driven.

3. Opportunistic plays (i.e., convert beta)

- Opportunistic plays following periods of asset dislocation / cyclical trough.
- Usually lasts for brief periods and is brought on by external market / technical factors (e.g., 2009, 2011).

A majority of our interview participants (managers) have been focussing more on relative value opportunities (versus arbitrage or opportunistic plays) in today's environment.

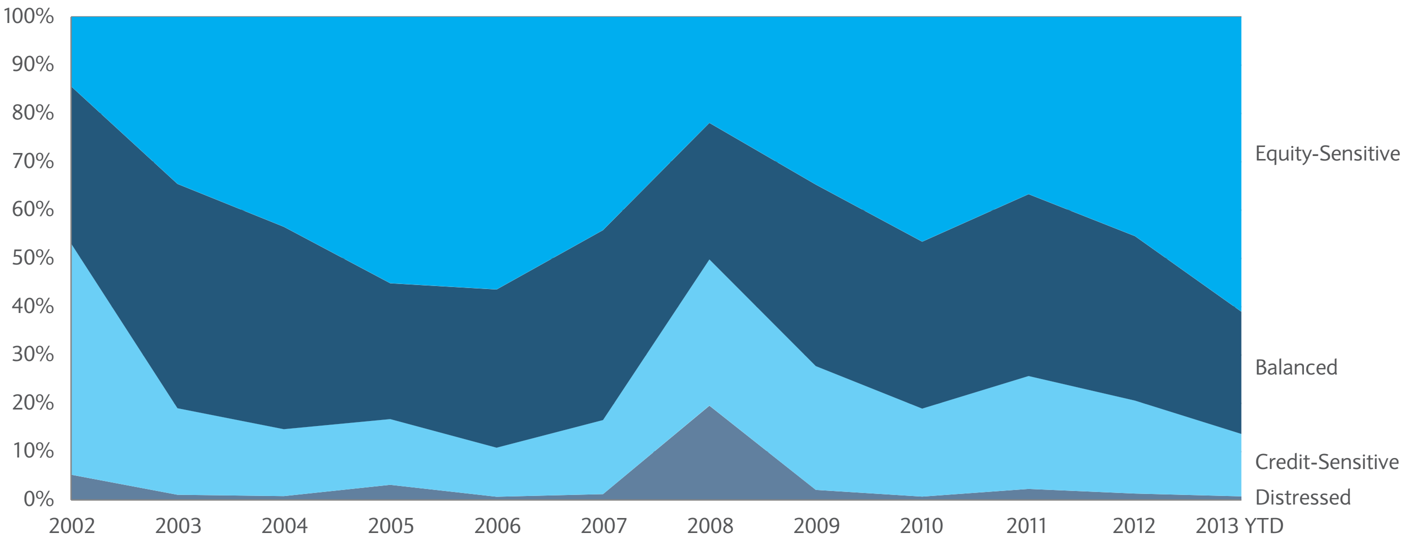
2. Common sub-strategies / trade types

Figure 10 depicts the typical strategy focus / trades employed at each stage of the convert profile. Key highlights:

- Equity-sensitive:
  - Focus on equities and event trades
  - Cheap puts and levered cash flows
- Balanced:
  - Focus on relative value, volatility arbitrage and event
  - Combination of credit, equity and volatility trades
- Credit-sensitive:
  - Focus on credit fundamentals and event
  - Combination of cheap calls and credit spread plays
- Distressed
  - Securities trading significantly below par value
  - Focus on recovery values, credit and event

FIGURE 11: Changes in Convert HF Strategy (II / III)

Composition of Outstanding Converts by Profile[1]



1. Barclays Convertibles Research estimates

A common investor allocation hurdle is the belief that a convert HF strategy is a 'black-box', model-driven strategy focussed mostly on extracting volatility cheapness. As a result, convert HFs tend to focus on the 'balanced' portion of the convert profile. Figure 11 shows the composition of outstanding converts in the marketplace by profile since 2002. According to the managers we interviewed, the mix of strategies in their portfolios today broadly reflect the profile mix of outstanding converts in the market, i.e., ~40% 'equity-sensitive', ~35% 'balanced', and 25% 'credit-sensitive'. This relatively even distribution potentially suggests that convert HFs are more likely to maintain a diversified portfolio today than rely on any single sub-strategy.

3. Evolution of convert HF strategies

An interesting observation from Figure 11 is that the composition of the profile of outstanding converts has changed substantially over time, and with that, the underlying strategy focus employed by convert HFs has also evolved. Below we summarise some key highlights of convert HFs' strategies for the past 20 years[13]:

- Late 1990s – Equity play
  - Telecom drove issuance; converts often became a pure equity instrument.
  - 2000 – 2003 – Credit play
  - Post-internet bubble, many converts lost equity sensitivity and became credit instruments.
  - 2004 – 2007 –Combination of equity and credit
  - Credit tightened, equities rallied, all stars aligned for converts.
- 2008 – 2010 – Credit play
  - Credit-driven recovery post-financial crisis (especially for asset classes lower in the capital structure).
- 2011 – present – Equity / event play
  - Focussed on fundamentals, idiosyncratic opportunities

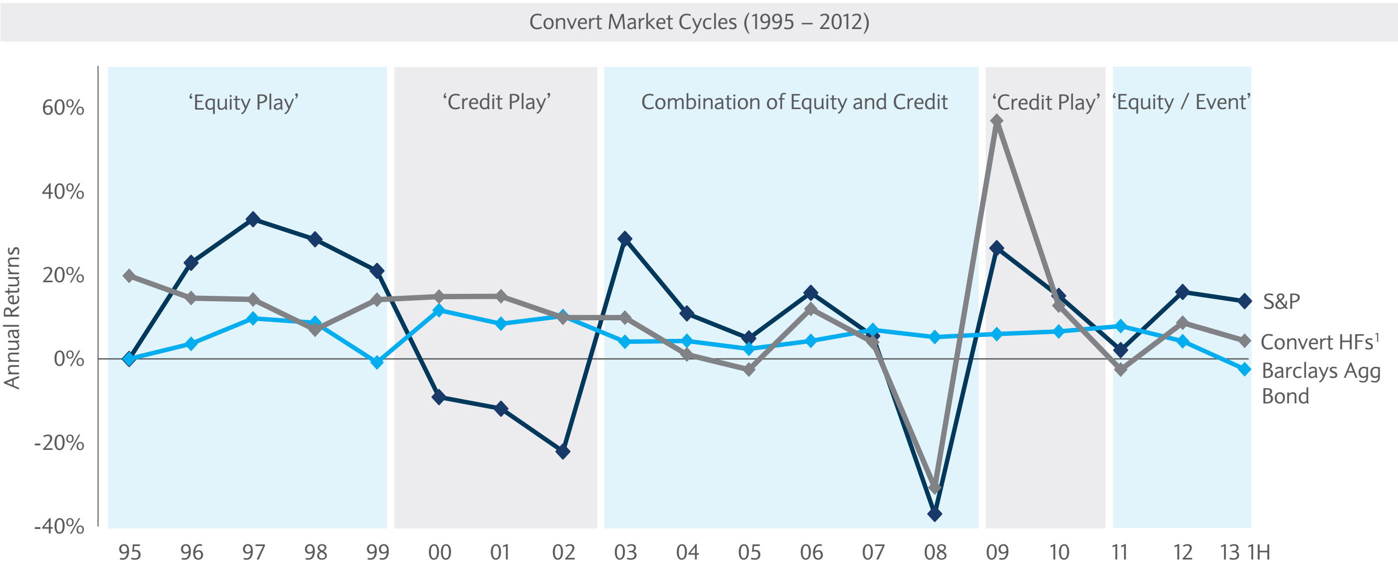
Because of converts' hybrid structure, convert HFs appear to have built-in flexibility to change strategy focus to try to achieve consistent returns. Some convert managers suggest that convert HFs, if managed well, can operate like 'mini-multi-strategy' HFs. This assertion goes against a common belief that convert HF strategy is a cyclical strategy that may not be attractive on a stand-alone basis across the market cycle. Figure 12 shows that, with the exception of 2008 and 2009, convert HFs have delivered fairly consistent performance on average, irrespective of market conditions.

Changes in talent base

The last topic we address in this section is change in the talent base dedicated to convert HFs. According to our study, a silver lining in the recent consolidation wave for convert HFs may be that surviving managers are more experienced. Following 2008, consolidation among convert HFs has eliminated some of the weakest players – several convert managers today are part of a battle-hardened group of survivors that has weathered multiple market cycles. Figure 13 shows that, on average, convert managers have a longer track record than other HF strategies at the end of Q3 2013 – 7.4 years versus an industry average of 6.3 years. This would imply that the average convert HF in existence today launched before the recent crisis (i.e., ~2005) versus the average FI RV / credit manager that has typically launched immediately after the crisis (2008).

[13] Strategic Consulting analysis

FIGURE 12: Changes in Convert HF Strategy (III / III)



1. Strategic Consulting estimate, average of BarclayHedge Convert Arbitrage index and HFR Convert Arbitrage index

Additionally, several of our interview participants have suggested that convert managers today tend to possess a more broad-based skill set than was the case historically. Many of the HFs that relied exclusively on a quantitative, 'black-box' approach focussed on extracting the 'cheapness' of convert securities have closed after 2008. A number of survivors today have experience / expertise in equities, volatility trading, as well as credit and event strategies. The combination of cross-asset class expertise, experience with derivatives, and an active trading mindset that convert managers need to have is probably a key reason why a disproportionate number of successful multi-strategy HFs today have a convert DNA (i.e., started their careers in HF investing with a focus on convert strategies).

V. Investment considerations

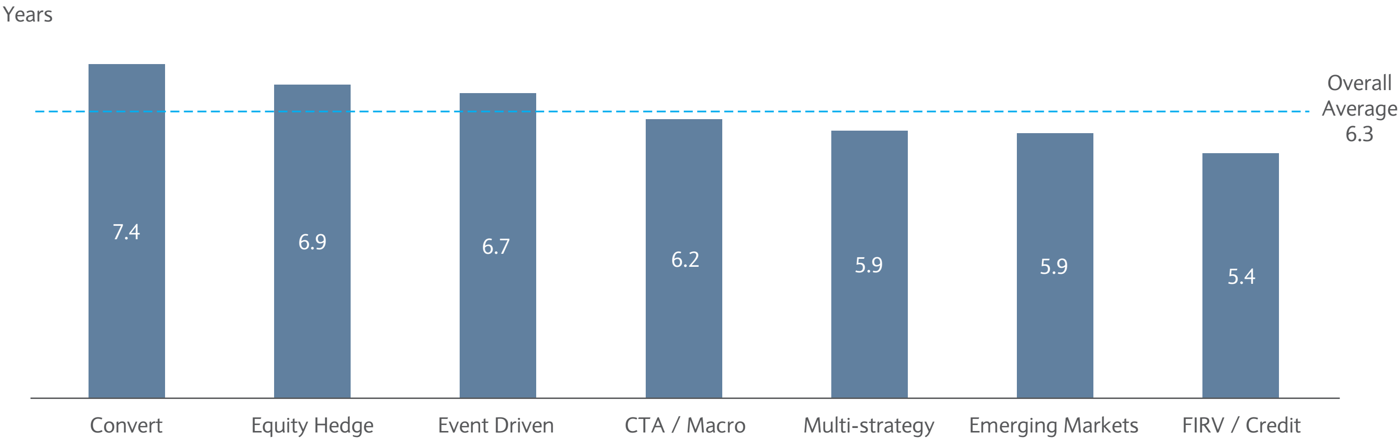
Allocation entry point

A concern some investors have expressed about allocating to convert HFs in the current environment is that the asset class has lately been 'expensive' (i.e., trading above theoretical valuations derived from current equity / bond prices, interest rate and volatility assumptions). Intrigued by this hypothesis, we examined the correlation between theoretical valuations and convert HF returns over time. Figure 14 shows the annual correlation between (1) end-of-month convert valuations[14] and (2) average returns over the subsequent three months for convert HFs[15]. If the investors' hypothesis were to be correct,

[14] Barclays Convertibles Research estimates [15] HFRI Convert Arbitrage Index

FIGURE 13: Convert HF Experience Base

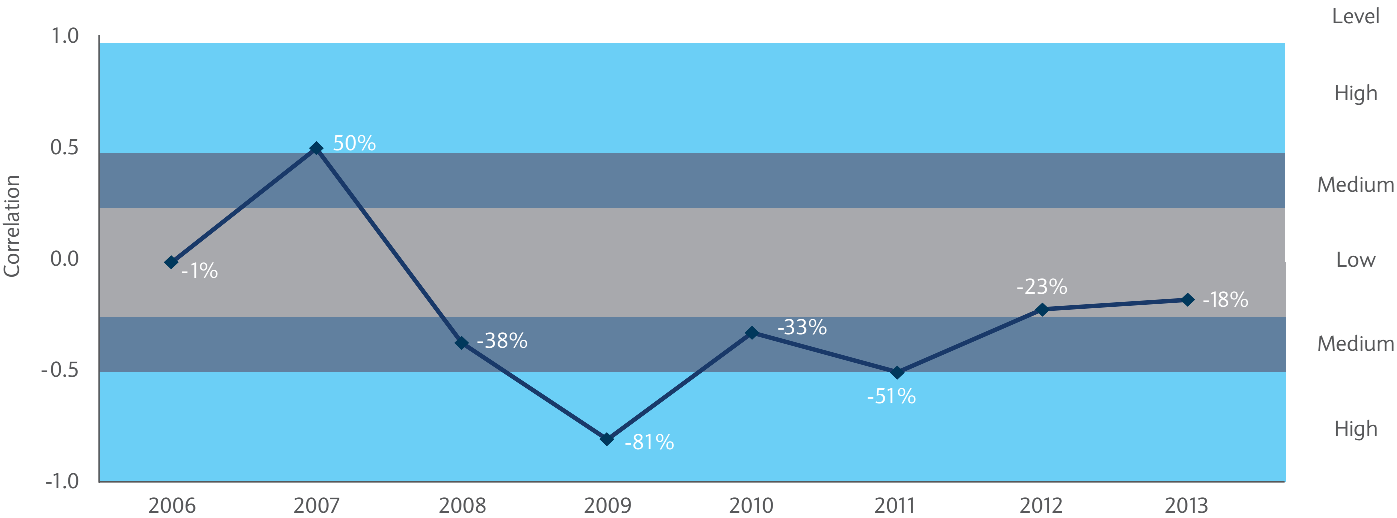
Average Age of Currently Active HFs by Strategy (as of 9/30/2013)[1]



1. HFR database

FIGURE 14: Valuation vs. Performance

Correlation between US Convert Valuation[1] and 3-Month Rolling Returns of HFRI Convertible Arbitrage Index[2]



1. Barclays Convertibles Research estimates 2. HFR

we would expect to see strong negative correlations between the two (i.e., lower valuations accompanied by higher returns). What we found instead is that correlations have been relatively weak (between -0.3 and 0.3) during 'normal markets' (e.g., 2010, 2012, 2013), and strong in times of 'extreme asset class cheapness' (e.g., 2009 and 2011). This is consistent with our earlier observation that returns in 'normal markets' tend to be driven by relative value opportunities versus the 'opportunistic' plays that dominate turbulent markets (i.e., convert beta, or asset class valuation).

Long-only versus HFs

A key question for investors contemplating converts is whether to access the asset class via a long-only or an HF product. Returns of long-only and convert HFs tend to be highly correlated (0.8+ over time[16]). However, the HF approach has historically generated a 'smoother' return profile, and a higher return on a risk-adjusted basis. Figure 15 shows annualised return (x-axis) and standard deviation (y-axis) of long-only and HFs convert funds over three, five and seven year time frames. Some key highlights:

- Over a three-year period, convert HFs captured ~70% of long-only convert returns (6.5% versus 9%) with less than half of the volatility (5% versus 11%).
- Over a longer time horizon (i.e., seven years), convert HFs delivered returns comparable to long-only converts (~6% annualised) with ~70% of the volatility.
- On a risk-adjusted basis, convert HFs have achieved similar or higher returns when compared to long-only convert managers across all three time periods tested.

Multi-strategy versus pure-play

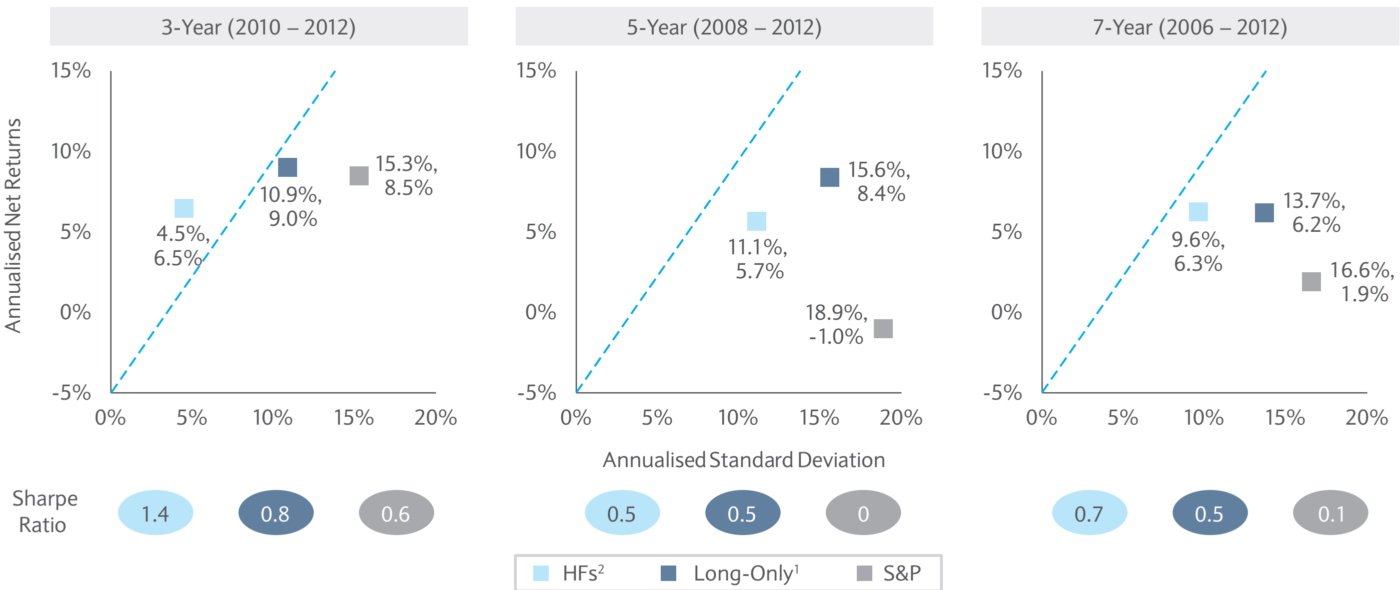
Investors have the choice of investing in convert HFs through either an allocation to a dedicated convert HF (pure-play) or to a multi-strategy fund that opportunistically allocates to converts. In this section, we discuss principal differences in the approach taken by these two types of HFs, along with key implications for investors:

Pure-play and MS HFs approach the convert opportunity differently. Pure-play convert HFs are arguably the specialists in the space who, on average, possess broader and deeper expertise in the asset class relative to MS HFs. They are also more likely to have an absolute return focus and are more likely to enter into all three types of trades discussed earlier (i.e., relative value, arbitrage and opportunistic). Select quotes from pure-play managers describing their edge:

- *"We are in the game – constantly in the flow. As a result, we see opportunities earlier than multi-strategy managers."*
- *"Multi-strategy HFs will get more involved as markets get better but we can take out the scalpel and take advantage of idiosyncratic trades."*
- *"Many MS HFs got into the market in 2009; we were already active in the market so we were able to move faster and further down the risk curve."*
- *"Market access is a key edge we have over MS. Pure-play convert funds are often invited to help price new deals and enjoy access to the limited capital and inventory of the dealers."*

[16] Strategic Consulting analysis using HFRI Convert Arbitrage index and US long-only convert returns

FIGURE 15: Performance: Long-Only vs. HFs



1. Barclays Convertibles Research estimates 2. Average of HFR, BarclayHedge convert index returns

MS HFs, on the other hand, have the flexibility to dial convert exposure up or down, based on the perceived opportunity set. Figure 16 shows the AUM allocated to converts by MS and pure-play convert HFs. As expected, MS allocation to converts has fluctuated substantially over time: most MS HFs tend to increase allocations right after sell-offs in the convert market, reflecting the fact that they approach converts as an 'opportunistic play'. For example, we estimate MS HFs' allocations to converts had reached ~$70bn at year-end 2009, accounting at that time for two-thirds of all the AUM allocated to converts (or ~20% of total AUM attributable to FI RV multi-strategy HFs). In contrast, MS HFs' allocations dropped to ~$20bn at year-end 2012 (or ~5% of FI RV MS AUM)[17], as the asset class became more fully priced. Select quotes from MS managers describing their edge:

- *"Converts tend to get richer / cheaper over the cycle – we can be opportunistic as we are not handcuffed to any one asset class or instrument type."*
- *"An MS approach allows us to better execute capital structure arbitrage as we are plugged into multiple markets."*
- *"There are advantages to pairing converts with other asset classes from a financing perspective."*

We think there are three key factors investors should consider before deciding which type of convert manager to allocate to:

1. Desired strategy exposure – MS HFs mostly focus on opportunistic plays (i.e., convert HF beta) versus pure-play firms that also strive to capture relative value and arbitrage opportunities (defined earlier). Additionally, for investors seeking a dedicated exposure to converts, 'pure-play' HFs may potentially be a more reliable avenue given their consistent allocation.
2. Risk tolerance – Broadly speaking, pure-play convert HFs tend to exhibit higher return / volatility than MS HFs. Pure-play firms may therefore be more attractive to investors with a higher tolerance for volatility.
3. Number of managers in investor HF portfolio – Investors on average, allocate 2 – 5% of their HF portfolio to converts. Therefore, on average, dedicated converts may make more sense to investors who allocate to 20+ HF managers.

The last question we explore on this topic is whether there is any performance differential between pure-play convert and MS HFs. In Figure 17, we show the median annual performance statistics for the two groups of funds taken from the HFR database:

1. Proxy used for pure-play HFs – Funds that list 'Convert Arb' as their primary strategy.
2. Proxy used for MS HFs – Funds that either list 'Convert Arb' as their secondary strategy or mention 'converts' in their strategy description. The primary strategy of these HFs may be MS, Credit, or Event Driven.

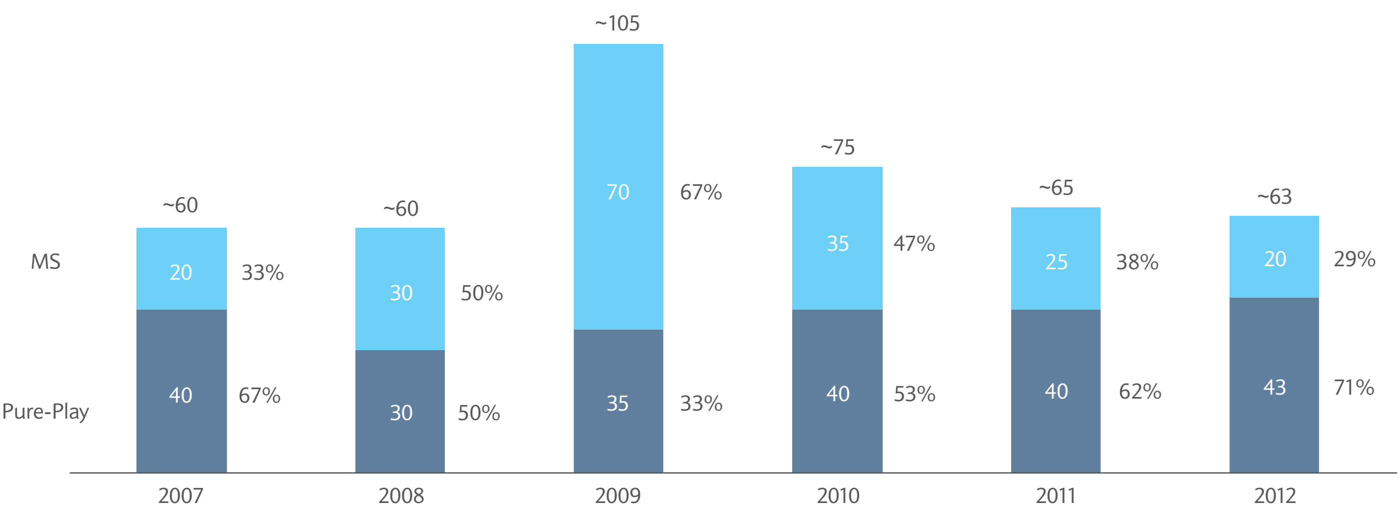
Since 2008, pure-play convert HFs appear to have demonstrated a performance edge over MS HFs in both absolute terms (30% versus 18% cumulative returns, respectively, over the 2008 – 2012 period) and on a risk-adjusted basis (average Sharpe ratio of 1.30 versus 1.12, respectively, over the same period). In particular

- During 'normal' markets such as 2010, 2012 and 2013, 'pure-play' convert HFs marginally outperformed their 'MS' counterparts.
- During market downturns such as 2008 and 2011, 'pure-play' convert HFs slightly underperformed MS HFs.

[17] Strategic Consulting convert market sizing model, incorporating factors such as convert market size, percent HF participation, average leverage levels, HFR RV Multi-strategy AUM and manager/investor interview feedback

FIGURE 16: 'Pure-Play' vs. 'MS' Convert AUM

AUM Allocated to Converts by 'Pure-Play' vs. 'MS' HFs ($bn)[1]



1. Strategic Consulting estimates; 'Pure-Play' AUM estimates based on AUM suggested by HFR Convertible Arbitrage index; 'MS' AUM estimated using Strategic Consulting proprietary market sizing model, incorporating convertible market size, percent of HF participation, average leverage levels, HFR FI RV Multi-strategy AUM, and manager / investor interview feedback

- Most of the pure-play firms' performance edge has been achieved historically during market recovery periods such as 2009. This may highlight the advantages of pure-play firms 'already being in the game' and therefore able to move more swiftly and aggressively when the opportunity arises.

In conclusion, dedicated convert HFs may offer an exciting opportunity to investors with a longer-term investment horizon (5 – 10 years), as they have generated strong Sharpe ratios over such periods historically, although an appetite for higher volatility (relative to MS HFs) is needed.

Manager selection

What attributes do investors look for in selecting a convert HF manager today? Generally speaking, investors in our sample prefer convert HFs that can demonstrate a broad skill set, fundamentally driven approach, and an active trading style:

- Multi-disciplinary skill set – There is a preference for managers who are able to demonstrate convert-specific expertise as well as experience in adjacent asset classes such as HY, credit and in event driven strategies (i.e., can demonstrate legal, documentation expertise).
- Fundamentally driven approach – Bias for managers with strong qualitative, fundamental capability / approach versus a systematic, 'black-box' approach.
- Opportunistic / active trading – Preference for managers with demonstrated ability and track record in navigating different market cycles, e.g., who know when to increase delta, under-hedge or over-hedge depending on their view. Investors also typically look for active trading in convert structures, which may require a greater level of involvement in the private debt / new issue markets.

On the flip side, investors tend to avoid managers whose strategy is solely focussed on extracting volatility cheapness and those with a volatile performance track record. On the business side, investors avoid those with a small (<$300mn) or rapidly declining asset base, as well as those without a strong infrastructure and financing arrangements (e.g., no multi-prime, no term financing).

VI. Conclusions

We think convert HFs may represent an interesting opportunity for some investors in the current environment for the following key reasons we discussed in this study:

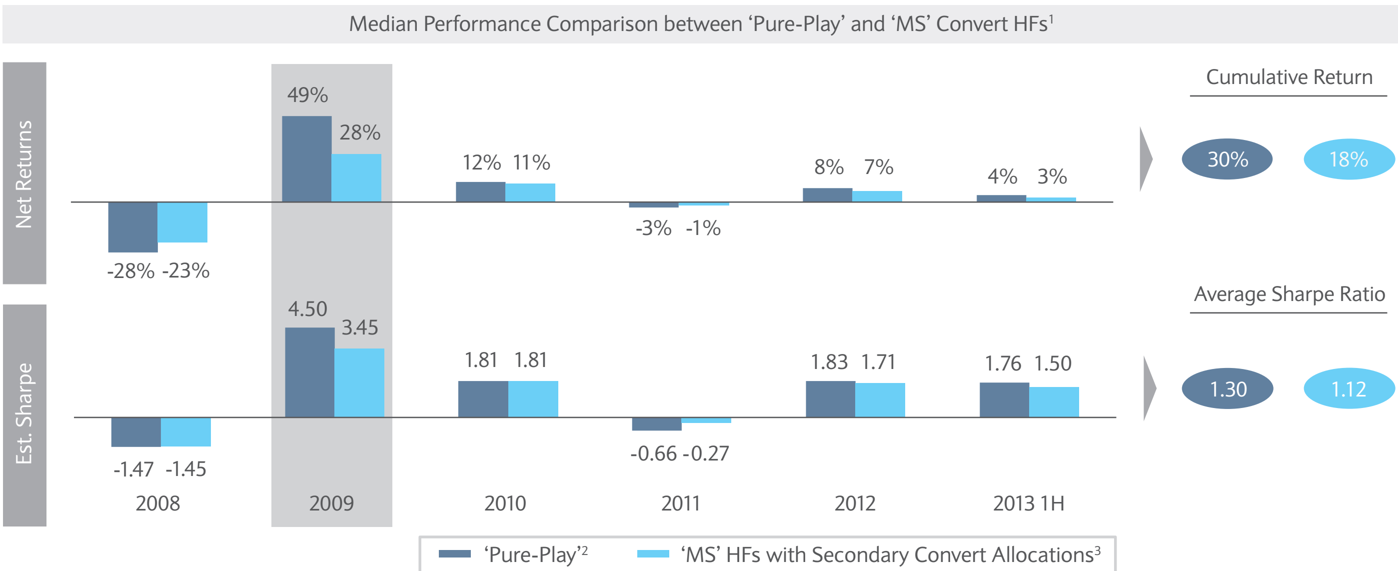
1. Asset class characteristics

- Built-in convexity of asset class with asymmetrical return profile
- Hybrid asset class that allows for multiple ways of investing. If executed right, could theoretically behave like an MS:
  - From an equity perspective, may allow better risk protection.
  - From a fixed income perspective, provides exposure to new companies.
  - From an arbitrage standpoint, multiple components embedded in one security facilitate arbitrage.

2. Changes in landscape

- A number of factors that contributed to the 2008 crisis have been addressed:
  - Concentration of players – Number of HF participants has declined both in absolute terms and versus outrights.
  - Use of leverage – Leverage today is nearly one-third of 2008 levels.
  - Less crowded trades – Convert strategy appears more broad based and fundamentally driven versus 2008.

FIGURE 17: 'Pure-Play' vs. 'MS' Performance



1. HFR database 2. Funds that list 'Convertible Arbitrage' as their primary strategy 3. Funds that either list 'Convert Arbitrage' as their secondary strategy or mention 'converts' in their strategy description. The primary strategy of these HFs may be MS, Credit, or Event Driven

- Impact of potential negative factors affecting convert HFs may be exaggerated in investors' minds:
  - Convert issuance has been at historical lows, however, that may be about to change.
  - The asset class, on average, is fully priced; however, we found that average theoretical valuation and convert returns are not strongly correlated.

3. Investment talent

- Surviving convert HFs seem to be more experienced and may have equity / volatility experience, and credit, and event-driven capabilities:
  - From a 'survival of the fittest' standpoint, many of the funds in existence today have weathered multiple market cycles.
- Based on our manager / investor interviews, survivors may have more multi-disciplinary, broad-based skill sets:
  - A number of successful HFs today have a convert DNA (e.g., Citadel, DE Shaw, and CQS).

4. Strategy neglected post-2008

- Investors, in general, have eschewed convert HFs since 2008. The strategy has shown much lower correlation between performance and net flows relative to other major HF strategies.

VII. Acknowledgments

The Strategic Consulting team would like to thank the managers and investors that participated in our interviews and made it possible for us to put this piece together. Per our promise to them, we have aggregated and presented the responses we received from them anonymously and in a way that no participant can be singled out. We want to thank them for allowing us to share this information with the broader universe of our readers.

VIII. Capital Solutions

The Capital Solutions team within Prime Services offers a unique blend of industry insights and tailored client solutions for a broad range of issues.

Capital Introductions

- Maintenance of ongoing investor dialogue to provide valuable feedback to HF managers.
- Introducing HF managers to a select number of interested investors.
- Hosting events that provide a forum for knowledge transfer and discussion / debate on industry issues that helps educate and inform both clients and investors.

Strategic Consulting

- Development of industry-leading content, driven by primary analysis, on the HF industry and its participants (e.g., HF and FoHF managers, institutional investors, investment consultants).
- Provision of management consulting services to HFs and asset managers on business topics such as the launch of a new strategy, marketing effectiveness, product development and organisational efficiency.
- Acting as an HF competence center internally for Barclays.

Contacts

Louis Molinari
Managing Director, Global Head of Capital Solutions
louis.molinari@barclays.com | +1 212 526 0742

Ermanno Dal Pont
Director, Head of Capital Solutions Europe
ermanno.dalpont@barclays.com | +44 20 3134 8649

Shelly Li
Vice President
shelly.li@barclays.com | +1 212 526 7657

Shagufah Nazaar
Assistant Vice President
shagufah.nazaar@barclays.com | +1 212 528 6348

Jennifer Bai
Analyst
jennifer.bai@barclays.com | +1 212 526 4655

Anurag Bhardwaj, CFA
Managing Director, Global Head of Strategic Consulting
anurag.bhardwaj@barclays.com | +1 212 526 5766

Roark Stahler
Director
roark.stahler@barclays.com | +1 212 526 9065

Tim Weaver
Vice President
tim.weaver@barclays.com | +1 212 526 7542

Freddie Parker
Analyst
freddie.parker@barclays.com | + 44 20 3134 5132

Disclaimer

BARCLAYS IS A MARKET PARTICIPANT. Barclays may act in several capacities (including issuer, market maker, index sponsor, swap counterparty, and calculation agent) simultaneously with respect to a product, giving rise to potential conflicts of interest which may impact the performance of a product.

Barclays, its affiliates and associated personnel may at any time acquire, hold or dispose of positions (including hedging and trading positions or soliciting investment banking or other services) which may impact the performance of a product.

THIS DOCUMENT IS PROVIDED FOR INFORMATION PURPOSES ONLY, IS SUBJECT TO CHANGE AND IS NOT BINDING. Any pricing included in this communication is indicative only. Barclays is not offering to sell or seeking offers to buy any product or enter into any transaction. Any transaction requires Barclays' subsequent formal agreement which will be subject to internal approvals and binding transaction documents.

Barclays is not responsible for the use made of this document other than the purpose for which it is intended, except to the extent this would be prohibited by law or regulation.

OBTAIN INDEPENDENT PROFESSIONAL ADVICE BEFORE INVESTING OR TRANSACTING. Barclays is not an advisor and will not provide any advice relating to a product. Before making an investment decision, investors should ensure they have sufficient information to ascertain the legal, financial, tax and regulatory consequences of an investment to enable them to make an informed investment decision.

Barclays is not responsible for information stated to be obtained or derived from third party sources or statistical services.

Any past or simulated past performance (including back-testing) contained herein is no indication as to future performance. Barclays makes no representation as to the accuracy, completeness or reasonableness of any performance data, modelling, back-testing or scenario analysis in these materials.

All opinions and estimates are given as of the date hereof and are subject to change. Barclays is not obligated to inform investors of any change to such opinions or estimates.

This document is being directed at persons who are professionals and is not intended for retail customer use.

For important regional disclosures you must read, click on the link relevant to your region. Please contact your Barclays representative if you are unable to access.

EMEA: www.barcap.com/emeadisclosures | APAC: www.barcap.com/apacdisclosures | U.S.: www.barcap.com/usdisclosures

IRS CIRCULAR 230 DISCLOSURE: Barclays does not provide tax advice. Please note that (i) any discussion of US tax matters contained in this communication (including any attachments) cannot be used by you for the purpose of avoiding tax penalties; (ii) this communication was written to support the Views addressed herein; and (iii) you should seek advice based on your particular circumstances from an independent tax advisor.

Barclays Capital Inc., the United States affiliate of Barclays Bank PLC, is responsible for the distribution of this material in the United States.

This document is confidential and must not be disclosed to third parties.

No part of this document may be reproduced, distributed or transmitted without Barclays' prior written permission.

Barclays Bank PLC offers premier investment banking products and services to its clients through Barclays Bank PLC. Barclays Bank PLC is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority and is a member of the London Stock Exchange. Barclays Bank PLC is registered in England No. 1026167 with its registered office at 1 Churchill Place, London E14 5HP.

Copyright Barclays Bank PLC, 2013 (all rights reserved).

P0408 | CSNY480364 A4 v2 | December 2013